

Literature status: the three-dimensional random-simplex monotonicity question

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Status. The focused open question in arXiv:1008.3944 is already answered by arXiv:1612.01893. This note records the exact identification; it does not claim a new result.

Source question. Luis Rademacher, *On the monotonicity of the expected volume of a random simplex*, states in the introduction that the dimension-three case was left open after proving monotonicity in dimensions one and two and disproving it in dimensions at least four. The Discussion (item 2, source PDF p. 16) asks verbatim:

“(3-D case) For Meckes’s strong conjecture, find an easy argument to disprove it for $d = 3$.”

Here the conjecture asserts that $K \subseteq L$ implies $\mathbb{E}V_K \leq \mathbb{E}V_L$ for the volume of the simplex spanned by four i.i.d. uniform points in a three-dimensional convex body.

Exact later answer. Stefan Kunis, Benjamin Reichenwallner, and Matthias Reitzner, *Monotonicity of the Sample Range of 3-D Data: Moments of Volumes of Random Tetrahedra*, arXiv:1612.01893, proves the following as Theorem 1 (supporting PDF p. 2).

Later theorem. *In $\mathbb{R}^3$ the expected volume of the sample range is in general not monotone in the underlying distribution: there are three-dimensional convex sets $L \subset K$ for which the expected volume of the tetrahedron generated by four uniform points is larger in L than in K .*

The strict inequality driving the construction is Proposition 1 (supporting PDF p. 3):

Decisive proposition. *For a volume-one tetrahedron T , the centroid c of one of its facets, and i.i.d. uniform $X_1, X_2, X_3, X_4 \in T$,*

$$\mathbb{E} |\text{conv}(X_1, X_2, X_3, c)| < \frac{13}{720} - \frac{\pi^2}{15015} = \mathbb{E} |\text{conv}(X_1, X_2, X_3, X_4)|.$$

Combined with Rademacher’s boundary-point characterization, this yields an infinitesimal convex-body variation that violates monotonicity.

Conclusion. The later theorem is precisely the negative answer requested in arXiv:1008.3944. The target is therefore classified as `literature_already_answered`; no independent solution packet is warranted.

References.

1. L. Rademacher, *On the monotonicity of the expected volume of a random simplex*, Mathematika 58 (2012), 77–91; arXiv:1008.3944.
2. S. Kunis, B. Reichenwallner, and M. Reitzner, *Monotonicity of the Sample Range of 3-D Data: Moments of Volumes of Random Tetrahedra*, arXiv:1612.01893 (2016).